

THEO KAWALISA PINEM

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ABOUT

Dedicated Computer Engineering student passionate and knowledgeable about IoT, AI and Machine Learning development, and Robotics. Eager to leverage strong academic knowledge and work ethic in challenging development and innovation projects.

EDUCATION

INSTITUT TEKNOLOGI SEPULUH NOPEMBER

2023 - Now

Bachelor of Computer Engineering

6th Semester

Current GPA : 3.51 / 4.00

TEFL : 530

PROJECTS

ADAPTIVE SMART IRRIGATION SYSTEM

Technologies: Raspberry Pi, ESP32, Python, C++, Machine Learning, IoT, Web Dashboard

- Engineered a distributed IoT architecture utilizing an ESP32 (C++) for real-time sensor data acquisition and a Raspberry Pi (Python) for central processing.
- Deployed a predictive machine learning model on the Raspberry Pi to autonomously optimize irrigation schedules based on environmental data telemetry.
- Developed and integrated a full-stack watering control system, enabling remote management and monitoring via an interactive dashboard.

AI-DRIVEN ELECTRICAL SAFETY AND LOAD CLASSIFICATION SYSTEM

Technologies: ESP32, PZEM-004T Sensor, Node-RED, Prometheus, Grafana, Python, K-Nearest Neighbors (KNN), MQTT

- Developed a real-time, ESP32-based electrical monitoring system utilizing PZEM-004T sensors and high-performance MQTT data transmission.
- Designed an automated data pipeline using Node-RED, Prometheus, and Grafana for seamless streaming and visualization of power metrics.
- Integrated a Python-based KNN machine learning model to classify electrical devices through custom power feature engineering.

SMART DRINKING WATER DISTRIBUTION SYSTEM

Technologies: ESP32, Load Cell, Flow Sensor, Ultrasonic Sensor, Flutter, Cloud Database, IoT

- Integrated load cells and flow sensors for high-precision, real-time tracking of water volume and consumption rates.
- Developed a cross-platform mobile application using Flutter to provide users with remote monitoring capabilities and hydration data management.
- Implemented an end-to-end IoT architecture to seamlessly synchronize physical sensor data with a cloud-based management system.

DISTANCE TARGET GAME

Technologies: ESP32, C++, Node.js, WebSockets, MySQL, Chart.js, Ultrasonic Sensor, HTML/CSS/JS

- Leveraged an ESP32 (C++) for high-precision physical distance measurement via an ultrasonic sensor, paired with a Node.js backend.
- Integrated WebSockets to enable synchronous, low-latency bidirectional communication between the hardware controller and the web interface.
- Designed a full-stack interactive game featuring a persistent leaderboard dashboard with visual analytics utilizing MySQL and Chart.js.

BIRD SHOOTER GAME

Technologies: ESP32, C++, LED Matrix, I2C LCD, Persistent Memory, Hardware Logic

- Programmed an ESP32 (C++) to command interactive game logic, integrate multiple hardware peripherals (LED Matrix, I2C LCD, Buzzer), and manage persistent high-score storage.
- Implemented dynamic difficulty scaling by mathematically optimizing hardware delays based on progressing player reaction scores.
- Designed a complete reaction-based arcade system delivering synchronized audio-visual feedback and real-time performance tracking.

FULLY AUTONOMOUS ESP32 ROBOTIC VEHICLE

Technologies: ESP32, C++, Ultrasonic Sensors, Motor Control Logic, Autonomous Navigation

- Engineered a fully autonomous robotic vehicle utilizing an ESP32 micro-controller for independent navigation and real-time decision making.
- Developed a "Smart Evasion" algorithm employing comparative sensor data to autonomously detect open paths and steer the vehicle into clear zones when trapped.
- Implemented a multi-sensor array system to provide comprehensive spatial awareness and blind-spot monitoring, ensuring zero-collision movement.

WORK EXPERIENCES

B401 ROBOTICS AND INTELLIGENT SYSTEMS LABORATORY

Coordinator of Telematics Workshop Practicum Assistant March 2026 - Now

- Leading the practicum program consisting of 70 participants and managing 11 staff.
- Directing technical training in soldering, circuit assembly, and PCB design.
- Supervising CAD modeling and prototyping to ensure design manufacturability.

Digital Circuit Practicum Assistant September 2025 - Now

- Teaching digital electronics and instructing in basic logic gates to around 60 participants.
- Guiding students in advanced digital design concepts.
- Mentoring teams on final projects of the practicum.

Laboratory Assistant September 2025 - Now

- Managing and maintaining laboratory equipment, calibration, and readiness.
- Supervising student experiments in robotics, embedded systems, and circuit design.
- Coordinated development and testing for robotics, IoT, and intelligent systems.

CODING CAMP POWERED BY DBS FOUNDATION X DICODING INDONESIA

Ai Engineer Cohort February 2026 - Now

- Selected from over 60,000 applicants nation-wide.
- Meeting program milestones through mentorship and consultations.
- Building an AI Capstone project using Deep Learning frameworks.

ORGANIZATIONAL EXPERIENCES

BADAN EKSEKUTIF MAHASISWA FAKULTAS TEKNOLOGI ELEKTRO DAN INFORMATIKA CERDAS

Expert Staff of Mentor in INCLINATION 2025 June 2025 - August 2025

- Directed, led, and taught a 10-person team of mentors.
- Supervised and monitored those mentors while they mentor 10 freshmen each.
- Provided strategic leadership to the mentor team and the freshmen.

Head of Development Bureau in Student Resource Department February 2026 - Now

- Supervising and leading a team consisting of 8 staff members.
- Directed and mentored the student development programs for incoming faculty students.
- Coordinated with the faculty deanery to report progress and ensure program alignment.

HIMPUNAN MAHASISWA TEKNIK KOMPUTER ITS

Staff of Student Resource Development March 2025 - December 2025

- Co-Led a major departmental introductory event (as Vice Head).
- Contributed to the strategic development of student resources with a 9-person team.
- Facilitated and assisted in training around 120 juniors.

PEMBINAAN KEROHANIAN MAHASISWA BARU KRISTEN ITS 2025

Steering Committee of Logistics

January 2025 - February 2026

- Conceptualized the entire program events for incoming christian freshmen.
- Acted as the key liaison with the ITS spiritual campus advisor with a 7-person team.
- Directly supervised and served as the Person-in-Charge (PIC) for the logistics division consisted of 14 persons.

MULTIMEDIA AND GAME EVENT

Head of Security and License Division

December 2024 - November 2025

- Directed and commanded a 12-person security and licensing team.
- Held ultimate accountability for event compliance and public safety.
- Provided strategic leadership and oversight across core functions.

CERTIFICATIONS

- Introduction to Software Engineering | RevoU
- Belajar Machine Learning untuk Pemula | Dicoding Indonesia
- Belajar Fundamental Deep Learning | Dicoding Indonesia